KENYA INSTITUTE OF CURRICULUM DEVELOPMENT
A Skilled and Ethical Society

DAFT

SENIOR SCHOOL CURRICULUM DESIGN

PHYSICS

GRADE 11

OCTOBER, 2025

First Published in 2025

All rights reserved. No part of this book may be reproduced, stored in a retrieval system or transcribed, in any form or by any means, electronic, mechanical, photocopy, recording or otherwise, without the prior written permission of the publisher.

ISBN: 978-9914-43-061-5

Published and printed by Kenya Institute of Curriculum Development

TABLE OF CONTENTS

NATIONAL GOALS OF EDUCATION ii
THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM iv
LEARNING OUTCOMES FOR SENIOR SCHOOL iv
TABLE 1: LIST OF SUBJECTS AT SENIOR SCHOOL v
TABLE 2: LESSON DISTRIBUTION AT SENIOR SCHOOL vi
ESSENCE STATEMENT vii
GENERAL LEARNING OUTCOMES vii
PROPOSED STRANDS viii
SUMMARY OF STRANDS AND SUB STRANDS ix
STRAND 1.0 MECHANICS AND THERMAL PHYSICS 1
STRAND 2.0 WAVES AND OPTICS 14
STRAND 3.0 ELECTRICITY AND MAGNETISM 22
STRAND 4.0 ENVIRONMENTAL AND SPACE PHYSICS 30
APPENDIX: LIST OF ASSESSMENT METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES 36

NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. **Foster nationalism and patriotism and promote national unity.**
   Kenya's people belong to different communities, races, and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect which enable them to live together in harmony and foster patriotism to make a positive contribution to the life of the nation.

2. **Promote the social, economic, technological, and industrial needs for national development.**
   Education should prepare the youth of the country to play an effective and productive role in the life of the nation.
   a) **Social Needs**
   Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution following the wake of rapid modernisation. Education should assist our youth in adapting to this change.
   b) **Economic Needs**
   Education in Kenya should produce citizens with the skills, knowledge, expertise, and personal qualities that are required to support a growing economy. Kenya is building up a modern and independent economy which needs an adequate and relevant domestic workforce.
   c) **Technological and Industrial Needs**
   Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognises the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills, and attitudes that will prepare our young people for these changing global trends.

3. **Promote individual development and self-fulfilment.**
   Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character.

4. **Promote sound moral and religious values.**
   Education should provide for the development of knowledge, skills, and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant, and integrated citizens.

5. **Promote social equity and responsibility.**
   Education should promote social equality and foster a sense of social responsibility within an education system that provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability, or geographical environment.

6. **Promote respect for and development of Kenya's rich and varied cultures.**
   Education should instill in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development to build a stable and modern society.

7. **Promote international consciousness and foster positive attitudes towards other nations.**
   Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights, and benefits that this membership entails.

8. **Promote positive attitudes towards good health and environmental protection.**
   Education should inculcate in young people the value of good health for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM

Senior School is the fourth level of Basic Education in the Competency Based Curriculum (CBC) that the learner shall proceed to after the Pre-Primary School, Primary School and Junior School levels. The essence of Senior School is to offer learners a pre-tertiary/university/pre-career experience in which the learners have an opportunity to choose pathways, tracks and subjects where they shall have demonstrated interest and/or potential at the earlier levels. Senior School comprises three years of education for learners generally in the 16-18 years age bracket and lays the foundation for further education and training at the tertiary level and the world of work. In the CBC vision, learners exiting this level are expected to be ***engaged, empowered and ethical citizens*** ready to participate in the socio-economic development of the nation.

At this level, the learner shall take seven (7) learning areas as recommended by the *Presidential Working Party on Educational Reforms* (PWPER, 2023). These shall comprise **four compulsory learning areas**, namely: English, Kiswahili, Essential Mathematics/Core Mathematics and Community Service Learning (CSL). Learners will choose between Core Mathematics (for learners pursuing the Science Technology Engineering and Mathematics Pathway) and Essential Mathematics (for learners pursuing the non-STEM pathways). The learner will select three learning areas depending on their career choice, aptitude, interest and personality with guidance by the Senior School career teacher.

LEARNING OUTCOMES FOR SENIOR SCHOOL

By the end of Senior School, the learner should be able to:

1. communicate effectively and utilise information and communication technology across varied contexts,
2. apply mathematical, logical and critical thinking skills for problem solving,
3. apply basic research and scientific skills to manipulate the environment and solve problems,
4. exploit individual talents for leisure, self-fulfilment, career growth, further education and training,
5. uphold national, moral and religious values and apply them in day-to-day life,
6. apply and promote health care strategies in day-to-day life,
7. protect, preserve and improve the environment for sustainability,
8. demonstrate active local and global citizenship for harmonious co-existence,
9. demonstrate appreciation of diversity in people and cultures,
10. manage pertinent and contemporary issues responsibly.

TABLE 1: LIST OF SUBJECTS AT SENIOR SCHOOL

Core Subjects	Arts & Sports Science	Social Sciences	Science, Technology, Engineering & Mathematics (STEM)
1. English 2. Kiswahili/KSL 3. Core Mathematics/ Essential Mathematics 4. Community Service Learning (CSL)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts	9. Literature in English 10. Indigenous Languages 11. Fasihi ya Kiswahili 12. Sign Language 13. Arabic 14. French 15. German 16. Mandarin Chinese 17. Christian Religious Education 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Biology 24. Chemistry 25. Physics 26. General Science 27. Agriculture 28. Computer Studies 29. Home Science 30. Aviation 31. Building Construction 32. Electricity 33. Metalwork 34. Power Mechanics 35. Woodwork 36. Media Technology 37. Marine and Fisheries Technology

Note:

a) Physical Education (PE) and Information Communication and Technology (ICT) will be offered to all learners to facilitate learning, enjoyment and acquisition of life skills.
b) Every school shall offer Pastoral/Religious Programme of Instruction (P/RPI) to enhance moral, spiritual and character development of the learners.
c) There shall be one (1) lesson for personal or group study to deepen knowledge and enhance skills in areas of choice by the learners.

Table 2 provides guidelines on distribution of lessons at Senior School.

TABLE 2: LESSON DISTRIBUTION AT SENIOR SCHOOL

Subjects	No of Lessons Per Week (40 minutes per lesson)
Core Subjects 4	
1. English	5
2. Kiswahili	5
3. Essential Mathematics/Core Mathematics	5
4. Community Service Learning	3
Elective Subjects	
5. Elective 1	5
6. Elective 2	5
7. Elective 3	5
* *Physical Education*	*3*
* *ICT Skills*	*2*
**PPI*	*1*
* *Learner Personal /Group Study*	*1*
Total Number of Lessons	**40**

ESSENCE STATEMENT

Physics is a body of knowledge exploring the theories, principles and laws that govern natural phenomena observed in the physical environment. The subject builds on the competencies introduced at Junior Secondary School level under the learning area of Integrated Science. In vision 2030 and sessional papers No. 1 of 2005 and No. 1 of 2019, the importance of science, technology and innovation has been prioritized for human capital development through education and training. Therefore, Physics forms an integral part of the STEM subjects in the Basic Education Curriculum Framework. It employs a scientific methodology of study with an emphasis on experimental approach to investigation, enhancing understanding of fundamental scientific concepts and principles. This leads to the acquisition of knowledge, skills, attitudes and values through precise and accurate scientific processes.

The subject provides the learner with opportunities to develop competencies by empowering them to be creative and innovative, leading to independent approaches in problem solving and management of their environment. It also prepares students for further training and the world of work by providing careers in STEM related pathways. The content will be anchored on Kolb's Experiential Learning Theory and Constructivism theory for teaching and learning. This theories state that learning is a continuous cycle involving concrete experience, reflection, conceptualization and experimentation, where learners gain knowledge by actively engaging with and reflecting on their experiences.

GENERAL LEARNING OUTCOMES

1. Relate Physics to technology and society to enhance the learner's appreciation of the physical environment.
2. Develop science process skills among learners through the use of appropriate instruments as they discover and explain the order of the physical environment.
3. Apply basic research and scientific skills to manipulate the environment and solve human problems.
4. Develop capacity for critical thinking through basic scientific skills and research in addressing pertinent & contemporary issues affecting the society.
5. Apply the principles of Physics for enhancement of innovations and entrepreneurial skills for development.

6. Use relevant skills and values to promote local and global citizenship for harmonious coexistence and appreciation of diversity in people.
7. Acquire adequate knowledge, skills, values and attitudes to enhance exploitation of individual talents for leisure, self-fulfilment, career growth, and for further education and training.
8. Apply acquired knowledge, skills, values, and attitudes for effective communication and utilization of information in technological advancement.

PROPOSED STRANDS

1. Mechanics and Thermal Physics
2. Waves and Optics
3. Electricity and Magnetism
4. Environmental and Space Physics

SUMMARY OF STRANDS AND SUB STRANDS

Strand	Sub Strand
1. Mechanics and Thermal Physics	1.1 Fluid flow
	1.2 Linear Motion
	1.3 Projectile motion
	1.4 Dimensional analysis
	1.5 Heat Transmission
2.0 Waves and Optics	2.1 Refraction of light
	2.2 Electromagnetic waves
	2.3 Thermionic emission and its applications
3.0 Electricity and Magnetism	3.1 Capacitors
	3.2 Magnetic effect of electric current
	3.3 Diodes
4.0 Environmental and Space Physics	4.1 Physics of wind formation
	4.2 Introduction to Rockets

STRAND 1.0 MECHANICS AND THERMAL PHYSICS

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Mechanics and Thermal Physics	**1.1 Fluid Flow** • Types of fluid flow (*steady flow, unsteady flow, laminar flow, turbulent flow, transitional flow, compressible flow, incompressible flow and viscous flow*). • Bernoulli's principle and its applications. • Equation of continuity. • Bernoulli's effect and its applications.	By the end of the sub strand, the learner should be able to: a) explain fluid flow as used in mechanics, b) describe types of fluid flow as used in mechanics, c) verify Bernoulli's principle as used in fluid flow, d) derive equation of continuity in fluid flow, e) describe the applications of Bernoulli's principle in real life situation, f) appreciate the applications of Bernoulli's principle in day-to-day life.	The learner is guided to: • discuss with peers the meaning and types of fluid flow, • carry out activities to demonstrate types of fluid flow, • use locally available materials to carry out activities to verify Bernoulli's Principle, • discuss with peers the stability of a vehicle in motion and safety demand it in relation to Bernoulli's Principle (*Risk Awareness and Management on road users; risky road behaviors such as speeding*), • work with peers to derive equation of continuity as used in fluid flow, • use print or non-print resources to search for information on verification of Bernoulli's Principle and the derivation of equation of continuity in fluid flow, • discuss with peers the applications of Bernoulli's principle in real life situation, • use print or non-print resources to search for information on applications of Bernoulli's principle in real life situations.	Why do birds and fish have a streamline shape?

Core competencies to be developed

- Communication and collaboration: the speaking and listening skills are developed as the learner speaks engagingly while discussing with peers the applications of Bernoulli's principle in real life situations.
- Learning to learn: the learner practices how to work with care and attention to details as they carry out activities to demonstrate types of fluid flow.

Values

- Respect: the learner learns to appreciate diverse opinions from peers as they discuss with peers the relationship between Bernoulli's Principle and the stability of a vehicle in motion.
- Peace: the learner shows care by avoiding hurting others while carrying out activities to demonstrate types of fluid flow.

Pertinent and Contemporary Issues

Socio-Economic and Environmental Issues.

- Safety and Security: the learner learns how to take precautions as they discuss with peers the stability of a vehicle in motion and safety around it in relation to Bernoulli's Principle.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Mechanics and Thermal Physics	**1.2 Linear Motion** • Terminologies used in linear motion (distance, displacement, speed, velocity, instantaneous velocity and speed, variable velocity, acceleration and rectilinear motion). • Motion-time graphs (distance – time, displacement – time, speed – time, *velocity – time graphs for bodies that are stationary, bodies moving with uniform speed/velocity and bodies with variable speed/velocity*). • Equations of uniformly accelerated motion.	By the end of the sub strand, the learner should be able to: a) explain the terminologies used in linear motion, b) determine speed and acceleration of a moving object, c) plot motion time graphs in linear motion, d) apply equations of uniformly accelerated motion to solve numerical tasks in linear motion, e) describe the applications of linear motion in real life situations, f) appreciate the applications of linear motion in day-to-day life.	The learner is guided to: • discuss with peers the terminologies used in linear motion, • use print or non-print resources to search for the meaning the terminologies used in linear motion, • perform experiments with peers to determine speed and acceleration of a moving object, • discuss with peers the need to observe road speed limits for safety of road users, • carry out activities leading to plotting and analysing motion time graphs, • use relevant quantities to derive equations of uniformly accelerated motion: $v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$ • use equations of uniformly accelerated motion to solve problems related to linear motion in day-to-day life of uniformly accelerated motion (shortest distance, braking force and overtaking speed), • use print or digital resources to search for information on motion time graphs, derivation and application of equations of uniformly accelerated motion in linear motion, • determine acceleration due to gravity using equations of linear motion, • discuss with peers applications of linear motion in real life situations.	1. Why is it important to observe road speed limits? 2. How can equations of uniformly accelerated motion be applied in day-to-day life?

Core competencies to be developed

- Digital Literacy: the learner develops digital manipulative skills as they use non print media to search for information on motion time graphs, derivation and application of equations of uniformly accelerated objects in linear motion.
- Critical thinking and problem solving: the learner explores different possible solutions as they use equations of uniformly accelerated motion to solve problems related to linear motion in day-to-day life of uniformly accelerated motion.

Values

- Patriotism: the learner practices following laid down procedure and regulations as they perform experiments with peers to determine speed and acceleration of a moving object.
- Integrity: the learner practices generating and using own data while carrying out activities to plot and analyzing motion time graphs.

Pertinent and Contemporary Issues:

Socio-Economic and Environmental Issues.

- Safety and Security: the learner learns the concepts of road safety and security as they discuss with peers the need to observe road speed limits for safety of road users.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Mechanics and Thermal Physics	**1.3 Projectile Motion** • Free fall. • Projectile motion (Vertical and horizontal projection). • Trajectory motion. • Acceleration due to gravity.	By the end of the sub strand, the learner should be able to: a) explain gravity as used in physics, b) demonstrate free fall of an object due to gravity, c) illustrate vertical and horizontal projectile motions due to gravity, d) determine acceleration due to gravity, e) illustrate the trajectory motion of bodies, f) appreciate the importance of acceleration due to gravity in day-to-day life.	The learner is guided to: • brainstorm on the meaning of gravity as used in physics, • carry out activities to demonstrate free fall of an object due to gravity, • discuss with peers the effects of free fall on the speed of a car moving down the inclined plane, • carry out activities to demonstrate vertical and horizontal projectile motion, of an object due to gravity, • perform an experiment to determine the acceleration of an object due to gravity, • use print or non-print resources to search for information on vertical projectile motion, horizontal projectile motion and free fall of an object due to gravity, • carry out activities to illustrate the trajectory motion of bodies, • discuss with peers the importance of projectile motion in real life experience.	Why does an object thrown upwards from earth's surface fall back to the ground?

Core competencies to be developed

- Critical Thinking and Problem Solving: the learner explores different possible solutions to a problem as they discuss the effects of free fall on the speed of a car moving down the inclined plane.
- Creativity and imagination: the learner combines different concepts to generate new ideas as they creatively as they carry out activities to illustrate the trajectory motion of bodies.

Values

- Integrity: the learner practices the use of resources sparingly as they carry out activities to demonstrate the free fall of an object due to gravity.
- Responsibility: the learner plays different roles as they carry out activities to illustrate the trajectory motion of bodies.

Pertinent and Contemporary Issues

Socio-economic issues

- Safety: the learner observes safety as they carry out activities to demonstrate free fall of an object due to gravity.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Question(s)
1.0 Mechanics and Thermal Physics	**1.4 Dimensional analysis** • Confirming the correctness of formulae and units involving mass, length, time, electric current and temperature.	By the end of the sub strand, the learner should be able to: a) describe the meaning of dimensional analysis as used in physics, b) verify the correctness of formulae using basic units in physics, c) appreciate the need of dimensional analysis in formulae and units.	The learner is guided to: • use print or non-print resources to search for the meaning of dimensional analysis as used in physics, • discuss with peers the meaning of dimensional analysis as used in physics, • use mass, length, time, electric current, temperature and their units to verify the correctness of basic physics-related formulae, • discuss with peers the need of dimensional analysis in formulae and units	Why should dimensional analysis be conducted before using formulae?

Core competencies to be developed:

- Self-Efficacy: as the learner develops confidence in using the common physics-related formulae as they use mass, length, time, electric current and temperature to verify the correctness of the formulae.
- Critical thinking and problem solving: the learner practices analysis of formulae and reasoning to validate an equation-related task as they verify the correctness of basic physics-related formulae.

Values

Respect: the learner learns to respect diverse opinions as they discuss with peers the need of dimensional analysis in formulae and units of basic quantities.

Pertinent and Contemporary Issues

Citizenship education

- Social Cohesion: the learner appreciates the globally used formulas in physics as they verify the correctness of basic physics-related formulae.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Question(s)
1.0 Mechanics and Thermal Physics	**1.5 Heat Transmission** • Heat. • Modes of heat transmission (conduction, radiation, and convection). • Applications of heat transmission.	By the end of the sub strand, the learner should be able to: a) explain heat as used in thermal physics, b) describe modes of heat transmission in matter, c) demonstrate modes of heat transmission in matter, d) describe the applications of various modes of heat transmission in day-to-day life, e) appreciate the applications of various modes of heat transmission in day-to-day life.	The learner is guided to: • discuss with peers the meaning of heat as used in thermal physics, • carry out activities to demonstrate the modes of heat transmission in matter, • discuss with peers the applications of various modes of heat transmission in real-life experience, • use print or non-print resources to search for more information on the applications of heat transmission in real life experience.	1. How is heat transmitted in solid, liquid, and gas? 2. How does a vacuum flask maintain the temperature of a liquid?

Core competencies to be developed:

- Imagination and Creativity: the learner generates ideas on how to handle objects at different temperatures as they discuss with peers the applications of various modes of heat transmission in real-life experience.
- Digital literacy: the learner develops digital manipulative skills while interacting with technology as they use non-print media to search for more information on the applications of heat transmission in real life experience.

Values

- Respect: the learner practices how to value diverse opinions of peers as they discuss the applications of various modes of heat transmission in real-life experience.
- Responsibility: the learner engages in assigned roles and duties as they carry out activities to demonstrate the modes of heat transmission in matter.

Pertinent and Contemporary Issues (PCIs)

Socio-Economic and Environmental Issues: the learner develops safety and security skills while dealing with heat as they carry out activities to demonstrate the modes of heat transmission in matter.

ASSESSMENT RUBRIC (HOLISTIC)

Level / Indicator	Exceeds Expectation	Meets Expectation	Approaches Expectation	Below Expectation
Ability to describe types of fluid flow as used in mechanics.	Correctly describes all types of fluid flow as used in mechanics, giving relevant illustrations.	Correctly describes all types of fluid flow as used in mechanics.	Describes at most 5 types of fluid flow as used in mechanics.	Describes at most 2 types of fluid flow as used in mechanics.
Ability to explain the applications of Bernoulli's principle in real life situations.	Correctly and comprehensively explains the applications of Bernoulli's principle in real life situations.	Correctly explains the applications of Bernoulli's principle in real life situations.	Explains the applications of Bernoulli's principle in real life situations, leaving out some key words.	Explains the application of Bernoulli's principle in real-life situations, given hints.
Ability to apply equations of uniformly accelerated motion to solve numerical tasks in linear motion.	Correctly and consistently applies equations of uniformly accelerated motion to solve numerical tasks in linear motion.	Correctly applies all equations of uniformly accelerated motion to solve numerical tasks in linear motion.	Applies at most 2 equations of uniformly accelerated motion to solve numerical tasks in linear motion.	Applies 1 of the equations of uniformly accelerated motion to solve numerical tasks in linear motion.
Ability to determine acceleration due to gravity.	Correctly and consistently determines acceleration due to gravity.	Correctly determines acceleration due to gravity.	Partially determines acceleration due to gravity.	Determines acceleration due to gravity, with hints.
Ability to verify the correctness of formulae using basic units in physics.	Correctly and consistently verifies the correctness of all formulae using basic units in physics.	Correctly verifies the correctness of all formulae using basic units in physics.	Verifies the correctness of at most half of the formulae using basic units in physics.	Verifies the correctness of at most quarter of the formulae using basic units in physics.
Ability to demonstrate modes of heat transmission in matter.	Correctly demonstrates all modes of heat transmission in matter, giving illustrations over the same.	Correctly demonstrates all modes of heat transmission in matter.	Demonstrates at most two of the modes of heat transmission in matter.	Demonstrates at most one of the modes of heat transmission in matter.
Ability to describe applications of various modes of heat transmission in day-to-day life	Correctly describes at least four applications of various modes of heat transmission in day-to-day life.	Correctly describes at least three applications of various modes of heat transmission in day-to-day life.	Correctly describes two applications of various modes of heat transmission in day-to-day life.	Correctly describes at most one application of various modes of heat transmission in day-to-day life.

STRAND 2.0 WAVES AND OPTICS

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Waves and Optics	**2.1 Refraction of light** • Verification of laws of refraction of light. • Determination of refractive indices of materials. • Dispersion of white light. • Light filtration. • Total internal reflection, effects, and applications.	By the end of the sub strand, the learner should be able to: a) explain refraction of light as used in waves, b) verify the laws of refraction of light as used in light waves, c) determine refractive index of a material from the law of refraction, d) describe dispersion and filtration of white light, e) illustrate total internal reflection of light in real-life situations, f) appreciate the applications of total internal reflection of light in day-to-day life.	The learner is guided to: • discuss with peers the meaning of refraction of light, • perform experimental activities to verify refraction of light using locally available materials, • carry out experimental activities to determine refractive indices of materials using locally available materials, • derive the mathematical formulas of refractive index from the law of refraction, • discuss with peers the dispersion and filtration of white light, • work with peers to illustrate the total internal reflection of light in real-life situations, • discuss with peers the effects and applications of total internal reflection, • use print or non-print resources to search for more information on the applications of total internal reflection.	1. How is knowledge of refraction of light important in day-to-day life? 2. How is the rainbow formed?

Core competencies to be developed:

- Communication and collaboration: the learner acquires good communication skills as they discuss with peers the effects and applications of total internal reflection.
- Self-efficacy: the learner develops self-confidence in mathematical application concepts as they derive mathematical formulae of refractive index from the law of refraction.

Values

- Respect: the learner practices respecting diverse opinions as they discuss with peers the dispersion and filtration of white light.
- Integrity: the learner practices how to be accountable on the prudent utilization of available resources as they carry out experimental activities to determine refractive indices of materials using locally available materials.

Pertinent and Contemporary Issues

Socio economic and environmental issues: the learner develops safety and security as they carry out experimental activities to determine refractive indices of materials using locally available materials.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Waves and Optics	**2.2 Electromagnetic waves** • Production, properties, detection and applications of electromagnetic waves. • Hazards of E.M radiation and safety.	By the end of the sub strand, the learner should be able to: a) explain production and detection of electromagnetic waves, b) arrange electromagnetic spectrum in order of increasing frequencies, c) describe the properties of electromagnetic waves, d) describe the hazards of electromagnetic waves, e) describe the applications of electromagnetic waves in real life situation, f) appreciate the applications of electromagnetic waves in day- to- day life.	The learner is guided to: • discuss with others the production and detection of electromagnetic waves, • arrange electromagnetic spectrum in order of increasing frequencies, • describe the properties of electromagnetic waves, • determine the energy of the electromagnetic waves using mathematical formula, • discuss with peers the hazards of electromagnetic radiations, • discuss with peers the applications of electromagnetic waves in nature (*Radio waves, Microwaves, Infrared radiation, Visible light, Ultraviolet radiation, X-rays and Gamma rays*).	How is electromagnetic wave important in day-to-day life?

Core competencies

- Critical thinking and problem solving: the learner learns how to evaluate and make decisions on the suitable property of an electromagnetic wave as they arrange the electromagnetic spectrum in order of increasing frequencies of a wave.
- Learning to learn: the learner works in care and attention to details as they arrange the electromagnetic spectrum in order of increasing frequencies.

Values

- Respect: the learner practices open-mindedness with different opinions from peers as they discuss the applications of electromagnetic waves in nature.
- Social justice: the learner practices the act of giving equal opportunities to each other in contributing their views as they discuss with peers the hazards of electromagnetic waves.

Pertinent and Contemporary Issues

Health promotion issues

- Effects and preventive measures: the learner acquires skills on how to prevent electromagnetic waves related health issues as they discuss with peers the hazards of electromagnetic waves.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Question(s)
2.0 Waves and Optics	**2.3 Thermionic emission and its applications** • Production of cathode rays and their properties. • CRO and its applications. • Production and properties of X-rays. • Applications of X-rays. • Safety precautions	By the end of the sub strand, the learner should be able to: a) explain the production and properties of cathode rays in waves, b) use CRO to analyze the frequency and amplitude of a wave, c) describe the production and properties of X-rays, d) determine the energy of X-rays using mathematical formulae, e) explain safety precautions necessary when handling X-rays, f) appreciate the applications of X-rays and CRO in day-to-day life.	The learner is guided to: • discuss with peers production and properties of cathode rays in waves (Cathode ray tube, CRO), • carry out experiments to analyze the frequency and amplitude of a wave using CRO, • discuss with peers the production and properties of X-rays, • use equations $E = hf$ and $E = \frac{hc}{\lambda}$ to determine the energy of X-rays, • discuss with peers the safety precautions necessary when handling X-rays equipment, • use print or non-print resources to search for more information on applications of CRO and X-rays, and safety precautions necessary when handling X-rays.	How does a CRO and X-ray work?

Core competencies

- Communication and collaboration: the learner develops science language communication skills as they discuss with peers the necessary safety precautions when handling X-ray equipment.
- Digital Literacy: the learner acquires digital manipulative skills as they use digital devices to search for more information on applications of CRO and X-rays.

Values

- Respect: as the learner shows open-mindedness as they discuss with peers the production and properties of cathode rays in waves.
- Unity: the learner cooperates in achieving a common goal as they carry out experiments to analyze the frequency and amplitude of a wave using CRO.

Pertinent and Contemporary Issues

Socio-economic and environmental issues

Safety and security: the learner discusses with peers the safety precautions necessary when handling X-ray equipment.

ASSESSMENT RUBRIC (HOLISTIC)

Level / Indicator	Exceeds Expectation	Meets Expectation	Approaches Expectation	Below Expectation
Ability to explain the applications of total internal reflection of light in day-to-day life.	Correctly and comprehensively explains applications of total internal reflection of light in day-to-day life.	Correctly explains applications total internal reflection of light in day-to-day life	Partially explains the applications of total internal reflection of light in day-to-day life.	Explains the applications of total internal reflection of light in day-to-day life, with prompt.
Ability to describe the properties of electromagnetic waves.	Correctly and comprehensively describes the properties of electromagnetic waves.	Correctly describes the properties of electromagnetic waves.	Partially describes the properties of electromagnetic waves.	With prompt, describes the properties of electromagnetic waves.
Ability to describe the applications of electromagnetic waves in real life situation.	Correctly and comprehensively describes the applications of electromagnetic waves in real life situation.	Correctly describes the applications of electromagnetic waves in real life situation.	Partially describes the applications of electromagnetic waves in real life situation.	With prompt, describes the application of electromagnetic waves in real life situation.
Ability to describe the production and properties of cathode rays.	Correctly and comprehensively describes production and properties of cathode rays.	Correctly describes production and properties of cathode rays.	Partially describes production and properties of cathode rays.	With prompt, describes production and properties of cathode rays.
Ability to explain the production and applications of X-rays.	Correctly and comprehensively explains the production and applications of X-rays.	Correctly explains the production and applications of X-rays.	Partially explains the production and applications of X-rays.	With prompt, explains the production and applications of X-rays.

STRAND 3.0 ELECTRICITY AND MAGNETISM

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Electricity and Magnetism	**3.1 Capacitors** • Electric field patterns. • Force between charges. • Capacitors and capacitor networks (capacitor types, charge and discharge, capacitance). • Applications of capacitors.	By the end of the sub strand, the learner should be able to: a) explain terminologies associated with static charges, b) determine the amount of force between charges, c) describe types of capacitors with their respective applications, d) illustrate charging and discharging of a capacitor, e) determine capacitance of a capacitor and effective capacitance, f) appreciate the application of capacitors.	The learner is guided to: • search for meaning of terminologies associated with static charges using print or non-print resources (electric field, electric field patterns, capacitors and capacitance), • draw electric field patterns between charged particles, • apply Coulomb's law to determine the magnitude of force between charged particles, • discuss with peers types of capacitors with their respective applications, • perform experiments to illustrate charging and discharging a capacitor using graphs in determination of energy stored in a capacitor, • search from print or non-print resources the mathematical formulae used to determine capacitance and effective capacitance (factors affecting capacitance, capacitor networks), • use print or non-print resources to search and discuss with peers the applications of capacitors in real life situations.	What is the significance of capacitors in electrical appliances?

Core competencies to be developed

- Learning to learn: the learner learns new skills as they perform experiments to illustrate charging and discharging a capacitor using graphs in determination of energy stored in a capacitor.
- Creativity and imagination: the learner uses creativity in work context as they perform experiments to illustrate charging and discharging a capacitor using graphs in determination of energy stored in a capacitor.

Values

- Responsibility: the learner develops accountability as they take care of apparatus while carrying out an experiment to determine energy stored in a capacitor.
- Respect: as the learner shows open-mindedness as they appreciate diverse opinions of others while sketching electric fields.

Pertinent and Contemporary Issues

Socio-economic and environmental issues: the learner practices safety and security in the laboratories while carrying out an experiment to determine energy stored in a capacitor.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Electricity and Magnetism	**3.2 Magnetic effect of electric current** • Magnetization and demagnetization (domain theory of magnetism). • Magnetic field due to a current. • Electromagnets. • Motor effect. • Applications of magnetic effect of electric current	By the end of the sub strand, the learner should be able to: a) explain magnetization and demagnetization using domain theory of magnetism, b) investigate the magnetic effect of a current flowing through a conductor, c) describe the working of an electromagnet, d) describe the direction of force produced on a current carrying conductor in a magnetic field, e) explain applications of magnetic effect of an electric current in day-to-day life, f) appreciate the applications of magnetic effect of electric current in day-to-day life.	The learner is guided to: • perform experiments to demonstrate magnetization and demagnetization, • search using print or non-print resources to explain magnetization and demagnetization using domain theory of magnetism, • perform experiments to investigate the magnetic effect of a current flowing through a conductor, • construct a working electromagnet using locally available materials, • visit a nearby garage, observe the working of electromagnetic alternator and discuss the role it plays in functioning as a motor of a vehicle in line with safety of road users, • discuss with peers the factors affecting strength of an electromagnet, • carry out activities to describe the force produced on a current-carrying conductor within a magnetic field • search using print and non-print resources the applications of a magnetic effect of electric current (*electric bell, electric motor, magnetic relay, microphone, telephone receiver, loud speaker, moving coil meters, circuit breakers among others*).	How does a motor work?

Core competencies to be developed

- Communication and collaboration: the learner develops speaking and listening skills as they discuss with peers the factors affecting strength of an electromagnet.
- Critical thinking and problem solving: the learner practices the nature of evaluation and decision making to use the suitable material as they model a working electromagnet.

Values

- Social Justice: the learner practices sharing resources equitably as they model a working electromagnet.

- Unity: the learner cooperates with others in striving to achieve common goals when performing experiments to establish the relationship between electric current and magnetism.

Pertinent and Contemporary Issues

Socio-Economic Issues: the learner learns how to observe road safety as they observe the working of electromagnetic alternators and discuss the role it plays in functioning as a motor of a vehicle in line with safety of road users.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Electricity and Magnetism	**3.3 Diodes** • P-N Junction diodes and biasing. • P-N Junction characteristics • Types of diodes and their uses.	By the end of the sub strand, the learner should be able to: a) explain the formation and biasing process of a p-n junction diode, b) describe the characteristics of a p-n junction diode, c) explain the applications of p-n junction diode in day-to-day life, d) identify types of diodes used in electronics, e) describe the application of p-n junction diode in rectification, f) appreciate the application of diodes in daily lives.	The learner is guided to: • discuss with peers the formation of a p-n junction diode, • perform an experiment to illustrate the biasing of a p-n junction, • design and perform an experiment to show the current-voltage (I-V) characteristics of a p-n junction diode for forward and reverse biased modes, • work with peers in creation of a model to describe application of p-n junction diode in rectification process using locally available resources (*half-wave rectification and full-wave rectification*), • use print and non-print resources to search for information on different types of diodes and their applications in day-to-day life.	How have p-n junction diodes promoted the development of electronics?

Core competencies to be developed:

- Creativity and Imagination: the learner practices combining different concepts to generate ideas while designing and performing experiments to show the I-V characteristics of a p-n junction diode for forward and reverse biased modes.

Values

- Unity: the learner appreciates efforts of others while designing and performing an experiment to show the I-V characteristics of a p-n junction diode for forward and reverse biased modes.
- Responsibility: the learner takes care of apparatus used while performing experiments to illustrate the biasing of a p-n junction.

Pertinent and Contemporary Issues

Socio-Economic and Environmental Issues: the learner acquires manipulative skills in designing safety and security gadgets in the environment as they perform an experiment to illustrate the biasing of a p-n junction.

ASSESSMENT RUBRIC (HOLISTIC)

Level / Indicator	Exceeds Expectation	Meets Expectation	Approaches Expectation	Below Expectation
Ability to illustrate charging and discharging of capacitors.	Correctly and systematically illustrates charging and discharging of capacitors.	Correctly illustrates charging and discharging of capacitors.	Partially illustrates charging and discharging of capacitors.	Illustrates charging and discharging of capacitors, with prompt.
Ability to determine capacitance of a capacitor and effective capacitance.	Correctly and consistently determines capacitance of a capacitor and effective capacitance.	Correctly determines capacitance of a capacitor and effective capacitance.	Partially determines capacitance of a capacitor and effective capacitance.	Determines capacitance of a capacitor and effective capacitance, with hints.
Ability to explain the applications of magnetic effect of an electric current in day- to- day life.	Correctly and comprehensively explains the applications of magnetic effect of an electric current in day- to- day life.	Correctly explains the applications of magnetic effect of an electric current in day- to- day life.	Partially explains the applications of magnetic effect of an electric current in day- to- day life.	Explains the applications of magnetic effect of an electric current in day-to-day life, with hints.
Ability to describe the formation and biasing process of a p-n junction diode.	Correctly and comprehensively describes the formation and biasing process of a p-n junction diode.	Correctly describes the formation and biasing process of a p-n junction diode.	Partially describes the formation and biasing process of a p-n junction diode.	Describes the formation and biasing process of a p-n junction diode, with cues.

STRAND 4.0 ENVIRONMENTAL AND SPACE PHYSICS

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
4.0 Environmental and Space Physics	**4.1 Physics of Wind Formation** • Wind patterns (polar easterlies, westerlies, horse latitudes, trade winds and the doldrums). • Forces acting on air masses (Pressure gradient force, Coriolis force, Turbulent drag). • Cyclones and anticyclones.	By the end of the sub strand the learner should be able to: a) describe the types of wind patterns in the atmosphere, b) describe forces acting on air masses in the atmosphere, c) describe the process of wind formation in the environment, d) explain formation of Cyclones and anticyclones, e) describe the importance of wind formation in day-to-day life, f) appreciate the importance of wind formation in nature.	The learner is guided to: • use print or non-print resources to search for information on the types of wind patterns, • discuss with peers the types of wind patterns in atmosphere, • use print or non-print media to search for information on the forces acting on air masses in the atmosphere, • discuss with peers on how cyclones and anticyclones contribute to wind formation, • discuss with peers the importance of wind formation in nature, • discuss with peers the applications of wind formation in real life.	1. How is wind formed? 2. How is wind formation applied in real-life situations?

Core competencies to be developed:

- Digital literacy: the learner acquires digital manipulative skills as they use digital resources to search for information on the forces acting on air masses in the atmosphere.
- Communication and collaboration: the learner develops speaking skills as they speak engagingly while discussing with peers on how cyclones and anticyclones contribute to wind formation.

Values

Respect: as the learner is patient with others during discussions with peers the applications of wind formation in real life

Pertinent and Contemporary Issues (PCIs)

Socio-Economic and Environmental Issues: the learner acquires knowledge in prevention of global warming as they discuss with peers the importance of wind formation in nature.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
4.0 Environmental and Space Physics	**4.2 Introduction to Rocket Science** • Rocket • Main parts of a rocket. • The forces involved in space flight as used in science.	By the end of the sub strand, the learner should be able to: a) explain the term rocket as used in space physics, b) identify the main parts of a rocket as used in space physics, c) demonstrate the upward motion of a rocket model, d) explore the forces involved in space flight, e) describe the applications of rockets in nature, f) appreciate the applications of rockets in day-to-day life.	The learner is guided to: • brainstorm on the meaning of the term rocket, • use print or non-print resources to search for information on the main parts of a rocket and their functions, • discuss the functions of the main parts of a rocket, • carry out activities to design a rocket model using locally available materials, • carry out activities to demonstrate upward motion of a rocket model using locally available materials (include air balloon), • carry out activities to examine forces involved in space flight, • use print or non-print resources to search for information on applications of rockets in day-to-day life. • discuss with peers the applications of rockets in day-to-day life.	How do rockets overcome Earth's gravity to travel through space?

Core competencies to be developed

- Imagination and Creativity: the learner combines different concepts to generate ideas as they carry out activities to design a rocket model using locally available materials.
- Digital literacy: the learner acquires digital manipulative skills as they use digital resources to search for information on applications of rockets in day-to-day life.

Values

- Responsibility: the learner shows accountability in playing different roles as they carry out activities to demonstrate upward motion of a rocket using locally available materials.
- Unity: the learner appreciates each other's efforts as they carry out activities to design a rocket model and demonstrate its upward motion using locally available materials.

Pertinent and Contemporary Issues (PCIs):

Socio-Economic and Environmental Issues (safety and security): the learner appreciates the role of rockets for safety as they discuss with peers the applications of rockets and satellites.

ASSESSMENT RUBRIC (HOLISTIC)

Level / Indicator	Exceeds Expectation	Meets Expectation	Approaches Expectation	Below Expectation
Ability to describe the types of wind patterns in the atmosphere.	Correctly and comprehensively describes the types of wind patterns in the atmosphere.	Correctly describes the types of wind patterns in the atmosphere.	Partially describes the types of wind patterns in the atmosphere.	With hints, describes the types of wind patterns in the atmosphere.
Ability to describe forces acting on air masses in the atmosphere.	Correctly and comprehensively describes forces acting on air masses in the atmosphere.	Correctly describes forces acting on air masses in the atmosphere.	Partially describes forces acting on air masses in the atmosphere.	With hints, describes forces acting on air masses in the atmosphere.
Ability to describe the process of wind formation in the environment.	Correctly and comprehensively describes the process of wind formation in the environment.	Correctly describes the process of wind formation in the environment.	Partially describes the process of wind formation in the environment.	With hints, describes the process of wind formation in the environment.
Ability to demonstrate upward motion of a rocket model.	Correctly and innovatively demonstrates upward motion of a rocket model.	Correctly demonstrates upward motion of a rocket model.	Partially demonstrates upward motion of a rocket model.	Demonstrates upward motion of a rocket model, with hints.
Ability to explore the forces involved in space flight.	Correctly explores all forces involved in space flight, giving relevant illustrations.	Correctly explores all forces involved in space flight.	Explores two forces involved in space flight.	Explores at most one force involved in space flight.

APPENDIX: LIST OF ASSESSMENT METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES

Assessment Methods in Science	Learning Resources	Non-Formal Activities
• Reflections • Game Playing • Pre-Post Testing • Model Making • Explorations • Experiments • Investigations • Conventions, Conferences and Debates • Teacher Observations • Project • Journals • Portfolio • Oral or Aural Question(s)s • Learner's Profile • Written Tests • Anecdotal Records	• Laboratory Apparatus and Equipment • Textbooks • Models • Digital media (Radio and TV education programmes, Kenya Education Cloud and OERs) • Print media (charts, pictures, journals, magazines) • Digital Devices • Software • Recordings • Resource persons	• Visiting the science historical sites. • Using digital devices to conduct scientific research. • Organising walks to have live learning experiences. • Developing simple guidelines on how to identify and solve some community problems. • Conducting science document analysis. • Participating in talks by resource persons on science concepts. • Participating in science clubs and societies. • Attending and participating in science and engineering fairs. • Organising and participating in exchange programmes. • Making oral presentations and demonstrations on science issues.